Locard's Exchange Principle - Python Demonstration

Python Code:

```
import os
import shutil
import time

# Create an "original" piece of evidence
with open("evidence_original.txt", "w") as f:
    f.write("Case File #001 - Suspect device log")
orig_stat = os.stat("evidence_original.txt")
print("Original file - created (ctime):", time.ctime(orig_stat.st_ctime))
print("Original file - modified (mtime):",
time.ctime(orig_stat.st_mtime))

# Simulate an investigator copying the file
time.sleep(1)
shutil.copy2("evidence_original.txt", "evidence_copy.txt")
copy_stat = os.stat("evidence_copy.txt")
print("Copied file - created (ctime):", time.ctime(copy_stat.st_ctime))
print("Copied file - modified (mtime):", time.ctime(copy_stat.st_mtime))

print("Locard's Exchange Principle:")
print("The copy operation itself created a new")
print("ctime on the destination file - every interaction leaves a
trace.")
```

Sample Output:

```
Original file - created (ctime): Thu Jul 16 05:13:46 2026
Original file - modified (mtime): Thu Jul 16 05:13:46 2026

Copied file - created (ctime): Thu Jul 16 05:13:47 2026
Copied file - modified (mtime): Thu Jul 16 05:13:46 2026

Locard's Exchange Principle:
The copy operation itself created a new
ctime on the destination file - every interaction leaves a trace.
```

Explanation:

This program demonstrates Locard's Exchange Principle in digital forensics. It creates an original evidence file, records its metadata, then copies it. The copied file has a new creation time (ctime) while preserving the original modification time (mtime). This shows that even a simple copy operation leaves a trace on the destination system, illustrating that every interaction with digital evidence creates forensic artifacts.